Adaptive Q&A Assistant

Using LLM and Neural SSM

12/04/2025

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Overview

1. **Motivation & Problem Definition**
   Why adaptive response modes are needed.
2. **Baseline Modeling & Limitations**
   Text-only regime classifier and its shortcomings.
3. **Neural State-Space Model (SSM)**
   Modeling conversation dynamics with latent state updates.
4. **Regime Switching Mechanism**
   Predicting Explainer / Tutor / Concise based on state + message.
5. **Explicit Intent Overrides & Training Framework**
   Handling explicit keywords and training with a hybrid dataset.
6. **Hybrid Chatbot Architecture, Results**
   Regime-driven LLM responses and evaluation outcomes.
7. **Limitation and Improvements**

Motivation & Problem Definition

Motivation

- Traditional chatbots are stateless and respond independently
- Cannot track conversation phase or user learning needs
- Produce the same style even when context changes
  Users require different explanation styles at different moments

Problem Definition

- Goal: dynamically choose between **Explainer / Tutor / Concise**
- Depends on both explicit intent and dialog context
- Formal objective: $\mathbf{z_t = f(x_t, h_{t-1})}$
- Need a system that adapts across multi-turn conversations

Baseline Modeling & Limitations

Baseline Regime Classifier

- DistilBERT embedding of the *current* user message
- Linear classifier predicts: Explainer / Tutor / Concise
- No history or sequential modeling
- Assumes all intent is explicit and local

Baseline Chatbot

- Uses one fixed system prompt for all responses
- Produces the same style throughout the conversation
- Cannot adapt to task changes or user preference shifts

Limitations

- Fails when user intent is implicit
- Cannot track conversational flow or evolving needs
- Does not recognize transitions like “explain → guide → summarize”

Neural State-Space Model (SSM)

Neural SSM

- Maintains latent conversation state $\mathbf{h}_t$
- Update rule: $\mathbf{h}_t = f_\theta(\mathbf{h}_{t-1}, \mathbf{x}_t, \mathbf{z}_t)$
- Learns temporal patterns and dialog structure

Regime Controller

- Predicts mode using (x_t, h_{t-1})
- 0 = Explainer, 1 = Tutor, 2 = Concise
- Learns smooth transitions across dialog phases

Regime Switching Mechanism

The baseline fails when:

- The user's request does not explicitly state the desired style
- Context matters but is not captured
- User intent changes over the course of the conversation
- The system should respect natural phases: **long explanation → tutoring → summary**

Example failure:
Turn 4 → user says: *"I need a detailed explanation now."*

- Baseline only detects "detailed" inconsistently
- Often ignores the conversational context
- Produces the wrong response style

This motivates the need for a model that incorporates conversation history.

Data Training & Testing

Our project does **not** use an external dataset; instead, you programmatically generate a **synthetic conversation dataset** that is specifically designed to test:

1. **Keyword-based regime classification**
2. **Sequence-based regime classification**
3. **Hybrid classification (keyword overrides + sequence defaults)**

The goal is to evaluate whether **history-aware modeling (SSM)** improves regime prediction over a **traditional text-only classifier**, under conditions where conversation structure matters.

Data Training & Testing

Each sample is a **conversation** consisting of:

- **SEQ_LEN = 5 user messages**
- Each message is synthetically generated
- Each message is paired with a **regime label**:

Result 1:

Keyword-based regimes → Baseline ≈ SSM → SSM not needed when label depends only on text.

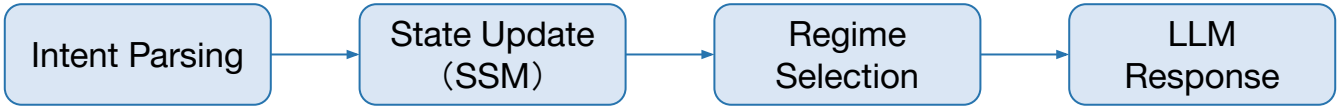
Result 2:

Sequence-based regimes → SSM >> Baseline → Sequence modeling adds real value.

Result 3:

Hybrid rules → SSM + overrides outperforms baseline → Your final chatbot respects both user intent and context.

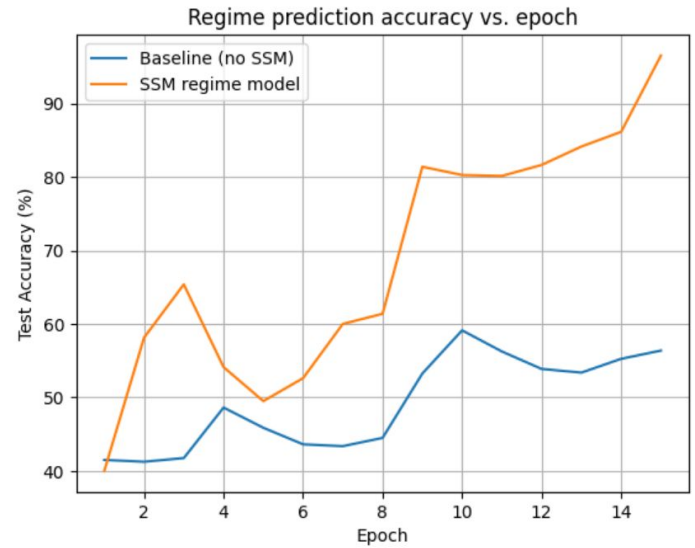
Chatbot Architecture & Results



Results

- Mode accuracy: ↑
- Stability across turns: ↑
- Example advantage:
  - Baseline jumps between modes

SSM chatbot maintains consistent Tutor mode over multi-round explanation

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Final baseline test accuracy: 56.37%
Final SSM test accuracy:      96.50%
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Areas of Improvement:

Limitations:

- Synthetic dataset is small
- SSM not trained end-to-end with LLM
- Regime classes are coarse
- No reinforcement learning from user feedback

Future directions:

- Train data on real dialogues
- More regimes (emotional tone, safety mode, beginner/expert)
- Model uncertainty
- Combine SSM with RNN-T or modern SSMs like Mamba
- Reinforcement tuning with human preferences

Thank You

Explicit Intent Overrides & Training Framework

Core Idea

- Maintain a latent state $\mathbf{h}_t$ representing conversation dynamics

Update Rule

- $\mathbf{h}_t = f_\theta(\mathbf{h}_{t-1}, \mathbf{x}_t, \mathbf{z}_t)$

Abilities

- Remembers earlier turns
- Learns structural patterns (e.g., early turns need long explanations)
- Supports sequence-dependent reasoning

Benefit

- Provides a principled way to model conversation flow over time